

WU SI

Ph.D. Candidate, The Hong Kong University of Science and Technology

Email: si.wu@connect.ust.hk Phone: (+86) 15698889290

EDUCATION

The Hong Kong University of Science and Technology , Ph.D.in Civil Engineering Research interests: HVAC system optimization; control-oriented modeling. Supervisors: Assistant Prof. WANG Zhe and Prof. CHEN Guanghao	Expected 2025
Shanghai Jiao Tong University , M.Sc. in Power Engineering and Thermophysics Master's thesis: <i>Numerical Investigations on a Turbofan Afterburner Fuel Pump</i> Supervisors: Prof. OUYANG Hua and Associate Prof. WU Yadong Academic Scholarship of SJTU	2019 - 2022
Dalian University of Technology , B.E. in Process Equipment and Control Engineering Outstanding Graduate of DUT Merit Student of DUT	2015 - 2019

PUBLICATIONS

1. **Wu, S.**, Zheng, W., Wang, Z.*, Chen, G.*, Yang, P., Yue, S.,... Wu, Y. (2025). AlphaDataCenterCooling: A virtual testbed for evaluating operational strategies in data center cooling plants. *Applied Energy*, 380, 125100. DOI: [10.1016/j.apenergy.2024.125100](https://doi.org/10.1016/j.apenergy.2024.125100)
2. **Wu, S.**, Yang, P., Chen, G., & Wang, Z.* (2025). Evaluating seasonal chiller performance using operational data. *Applied Energy*, 377, 124377. DOI: [10.1016/j.apenergy.2024.124377](https://doi.org/10.1016/j.apenergy.2024.124377)
3. **Wu, S.**, Wu, Y.*, Tian, J., & Ouyang, H. (2022). On the cavitation-induced collapse erosion of a turbofan fuel pump. *Engineering Applications of Computational Fluid Mechanics*, 16(1), 1048-1063. DOI: [10.1080/19942060.2022.2067243](https://doi.org/10.1080/19942060.2022.2067243)
4. Yang, Z., Ming, L., **Wu, S.**, Wu, Y., Tian, J., & Ouyang, H. (2022, June). On the Mode Characteristics of Rotating Instability with Different Tip Clearances. In *Turbo Expo: Power for Land, Sea, and Air (Vol. 86120, p. V10DT37A013)*. American Society of Mechanical Engineers. DOI: [10.1115/GT2022-82072](https://doi.org/10.1115/GT2022-82072)

RESEARCH EXPERIENCE

Model Predictive Control for Energy Efficient Data Center <i>School of Engineering, HKUST</i> Developed an open-sourced virtual testbed <i>AlphaDataCenterCooling</i> for data center cooling plant control strategy optimization. https://github.com/wfzheng/AlphaDataCenterCooling Proposed a method for benchmarking chiller performance using operational data without shutdown measurements.	Feb. 2023 - June 2024
On the Cavitation-Induced Collapse Erosion of a Turbofan Fuel Pump <i>School of Mechanical Engineering, SJTU</i> High-speed multi-phase flow field analysis based on computational fluid dynamics (CFD) simulations. Optimized design of a turbofan fuel pump based on fluid-structure coupling simulations.	Oct. 2020 - Aug. 2022
Design and Performance Analysis of an Aeration Fan <i>School of Energy and Power Engineering, DUT</i> Designed an impeller based on the specified operating conditions. Simulated the impeller's performance under these conditions using the CFD software NUMECA. Enhanced the internal flow characteristics of the impeller by adding splitter blades.	Dec. 2018 - May 2019
Structural Design of a Composite Material Pressure Vessel <i>School of Ocean Science and Technology, DUT</i> Designed the structure of a filament-wound pressure vessel. Investigated the impact of material yield strength and geometric parameters on the sealing performance of a metal sealing ring.	Dec. 2016 – Dec. 2017

SERVICE

Teaching Assistant Fundamental of Green Buildings
Energy System Modelling for Buildings and Cities